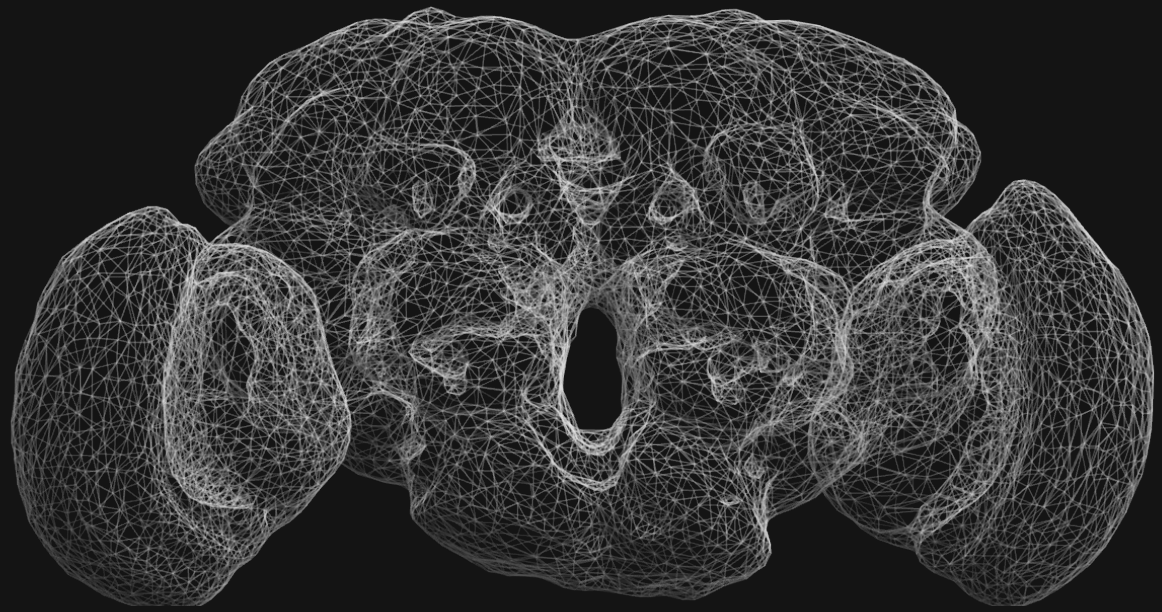
Understanding Thermal Nociception Behavior: Analysis of the Drosophila Connectome



Jessica Horton

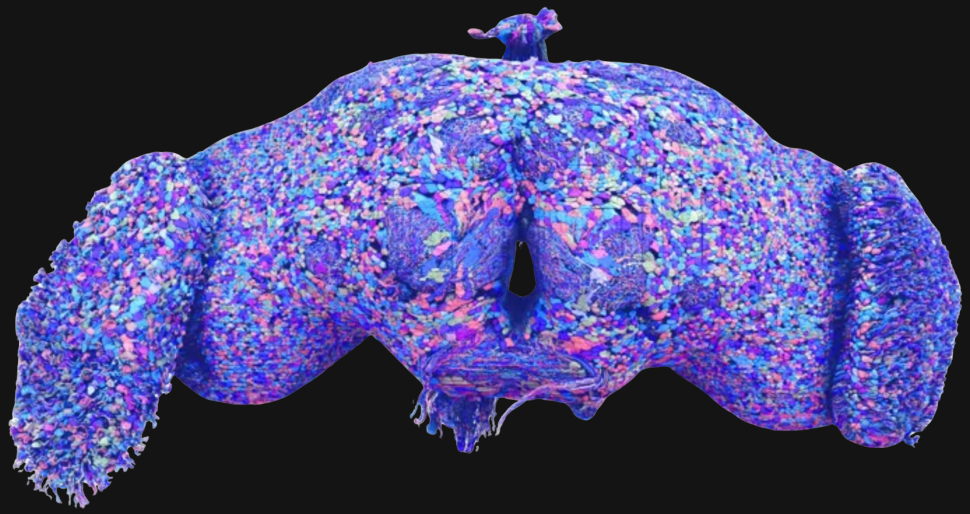
Background

- For over 100 years, *Drosophila melanogaster* have served as a wonderful model organism for scientists to study

- Low cost
- Large sample size
- Easy to maintain
- Genetic similarities to human beings

Yamaguchi *et al.*

History of the F.A.F.B.



2013: Dr. Davi Bock submerges an Female Adult Fly Brain (F.A.F.B.) into a hardening chemical solution

2018: His team has successfully sliced and imaged the brain using Electron Microscopy

2018: Scientists at Princeton's Neuroscience Institute train a large language model on the data and develop their first draft of the connectome

2024: The connectome is fully annotated and published

Connectome: computationally created neural connectivity map of a brain

Schlegel *et al.*, Zhang *et al.*

My Project

- Investigate two neurons known to be involved in temperature-induced pain processing due to their production of the neuropeptide, Allatostatin C

- What do the first order neuronal connections between these two neurons look like?

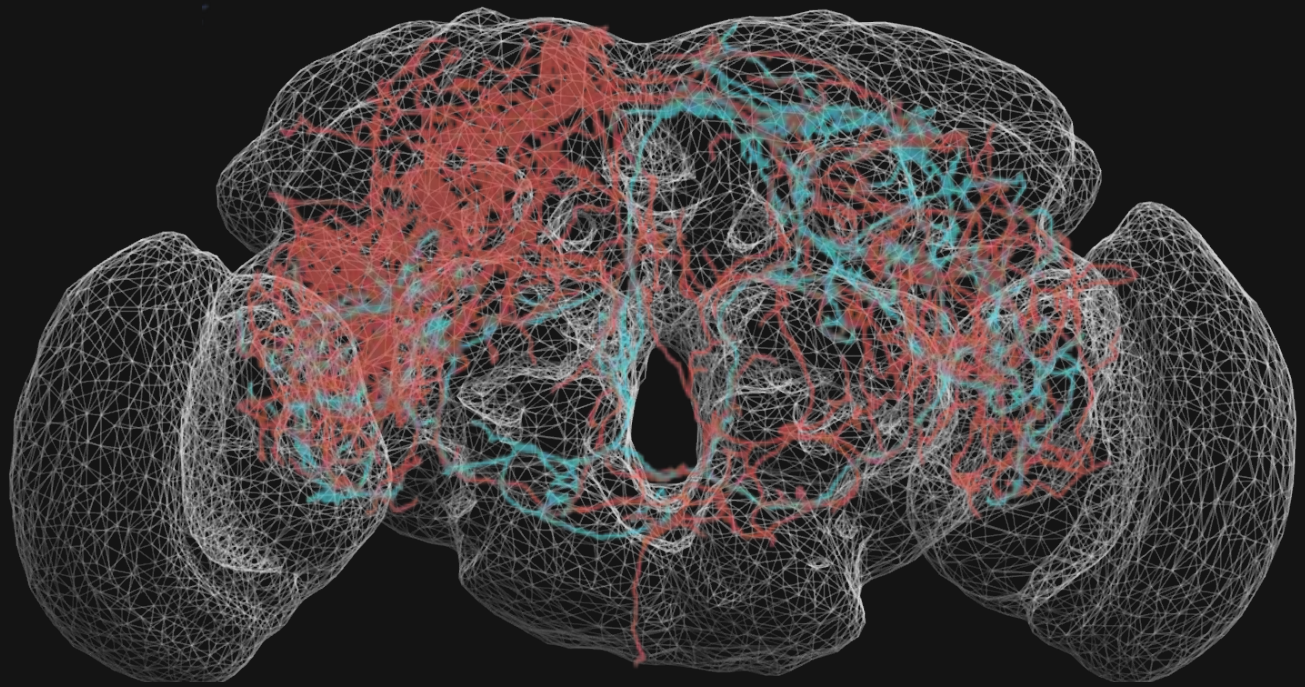
- Develop a 3-Dimensional visualization of the first order upstream, downstream, and reciprocal neurons

- Then, create a 3D graphic of the overlapping first order neurons shared between our two neurons of interest

Bachtel *et al.*

PLP.79 and PLP.LO.13

The two neurons known to possess Allatostatin C in the FAFB connectome



My Approach

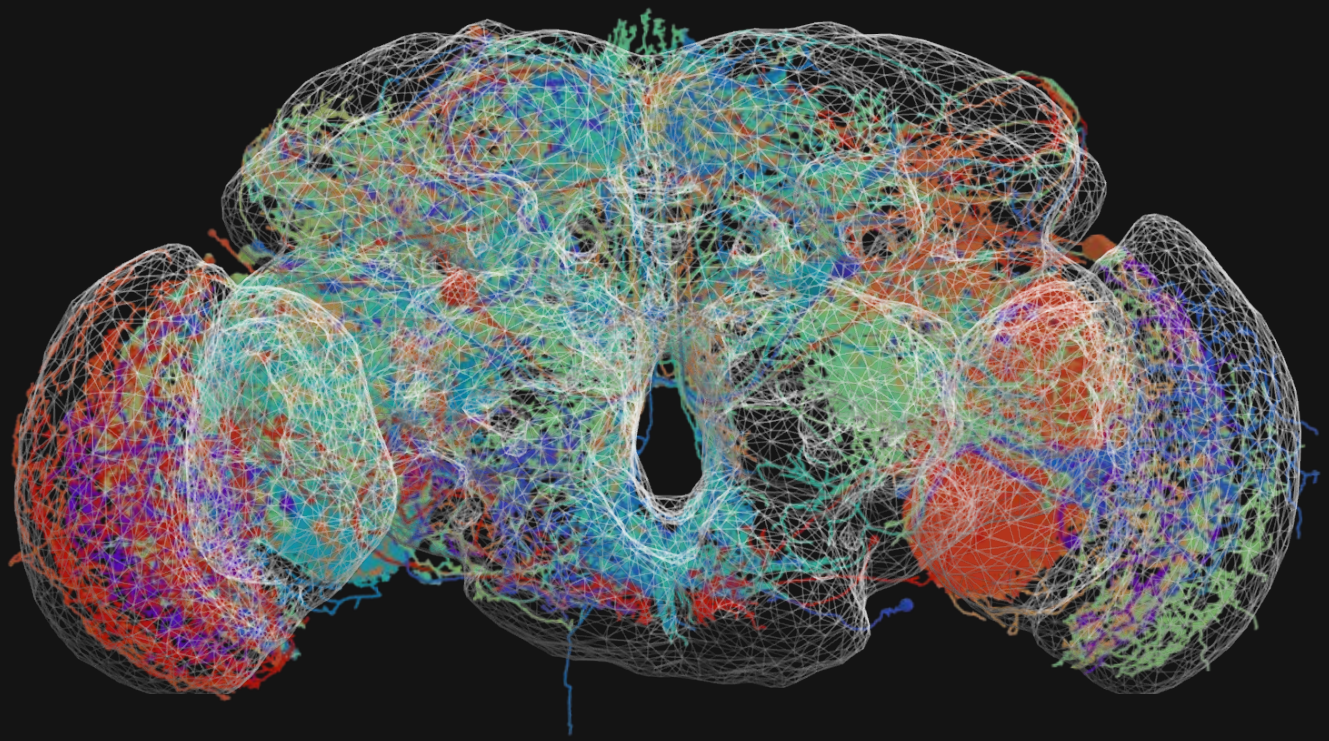
```python
def csv_to_swc(csv_location, swc_directory, empty_list):
    csv_file = pd.read_csv(csv_location, header = None)
    for i in range (0, csv_file.size):
        file_name = csv_file.iat[0, i]
        swc_file_path = f"{swc_directory}\\{file_name}.swc"
        swc_file = navis.read_swc(swc_file_path)
        empty_list.append(swc_file)
```

```python
def brain_graph(swc_list, colors):
    fig, ax = navis.plot3d([swc_list], radius = False, color = colors)
    return fig, ax
```

```
plp_79_upstream = []
# csv_to_swc(csv_location, swc_directory, empty_list)
csv_to_swc(r"C:\Users\jjess\Documents\MEGA\Bellemer_L
# brain_graph(swc_list, colors)
brain_graph(plp_79_upstream, "rainbow")
```

PLP.79 upstream

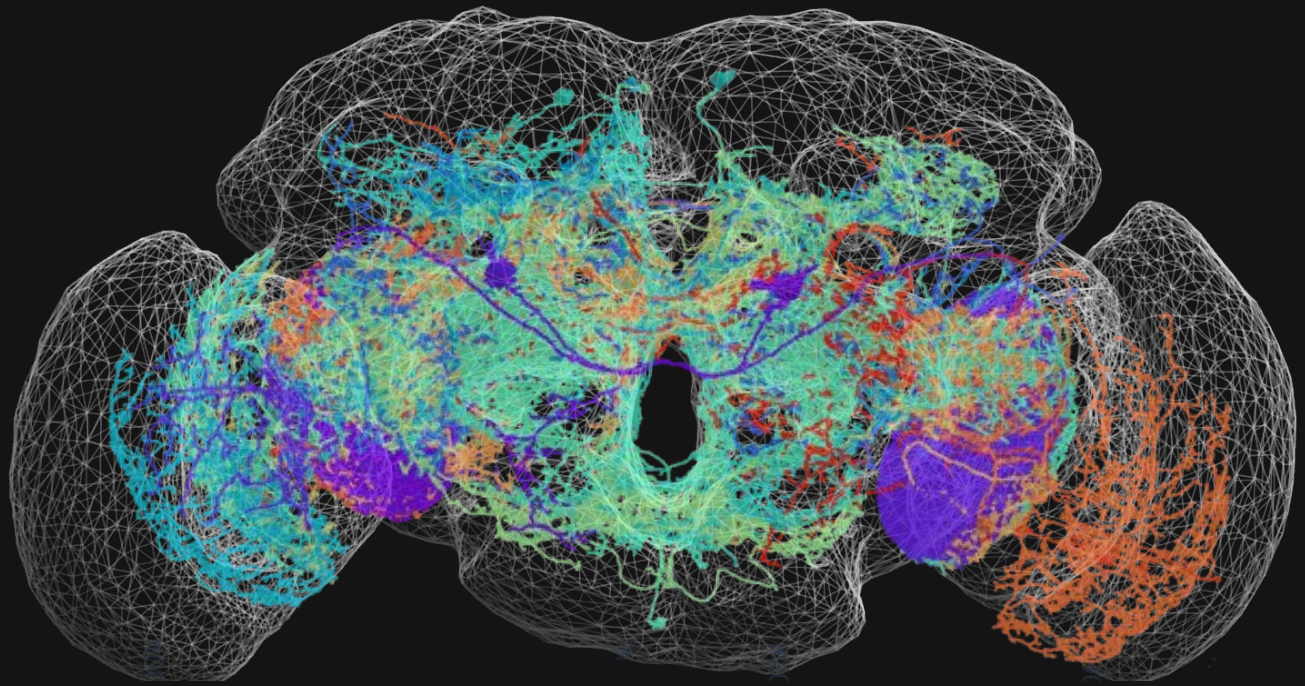
Neurons sending electrical activity into PLP.79



```
plp_79_downstream = []
# csv_to_swc(csv_location, swc_directory, empty_list)
csv_to_swc(r"C:\Users\jjess\Documents\MEGA\Bellemer_Lab
# brain_graph(swc_list, colors)
brain_graph(plp_79_downstream, "rainbow")
```

PLP.79 downstream

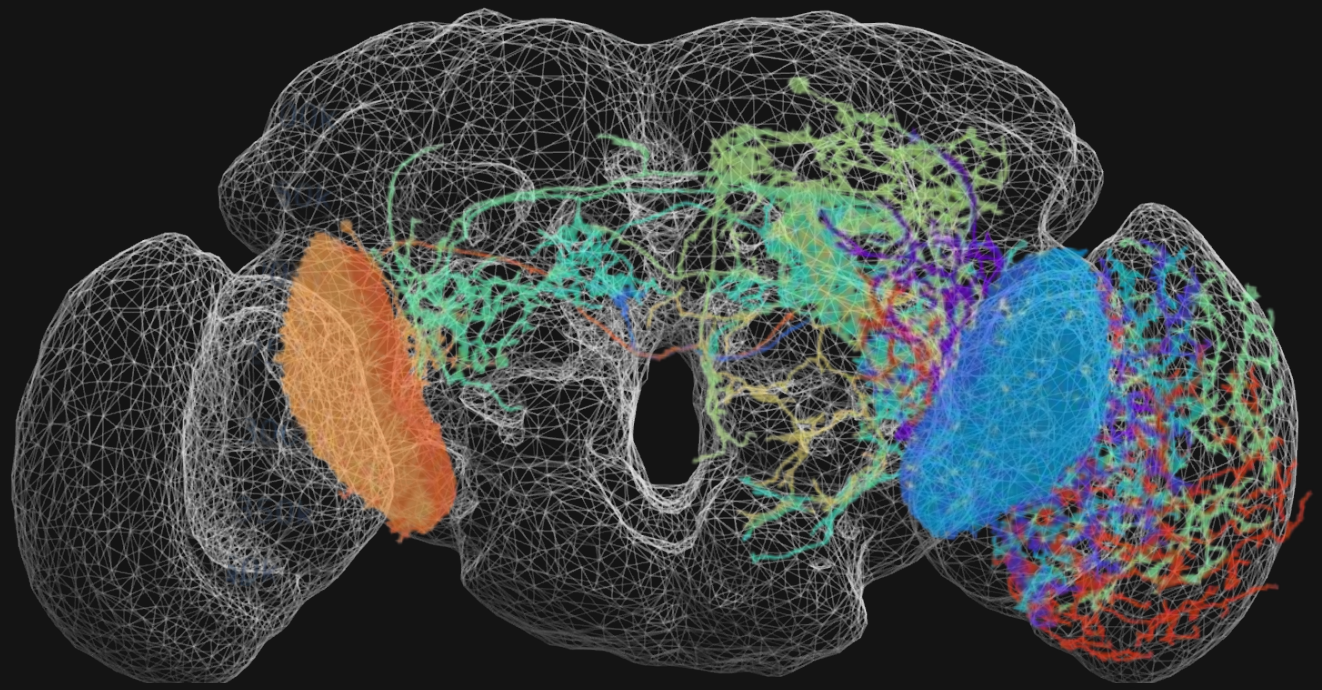
Neurons receiving electrical activity into PLP.79



```
plp_79_reciprocal = []
# csv_to_swc(csv_location, swc_directory, empty_li
csv_to_swc(r"C:\Users\jjess\Documents\MEGA\Bellemer
# brain_graph(swc_list, colors)
brain_graph(plp_79_reciprocal, "rainbow")
```

PLP.79 reciprocal

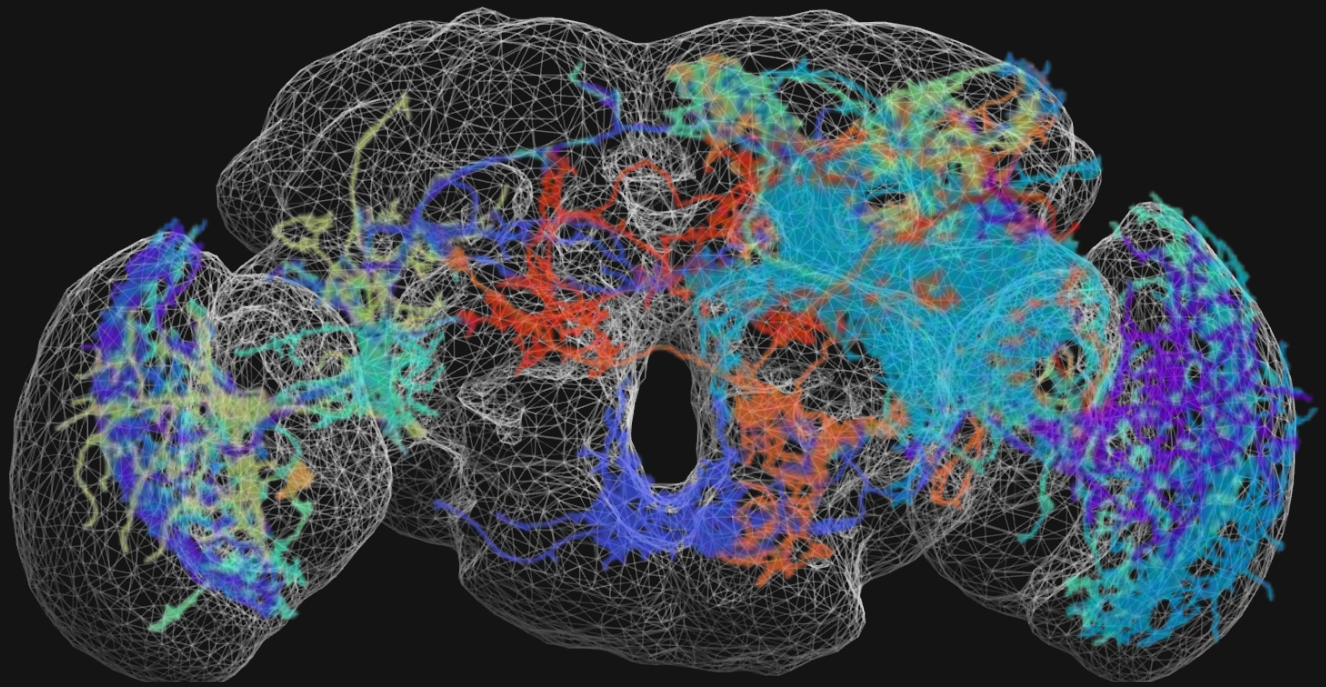
Neurons sending and receiving electrical activity to and from PLP.79



```
plp_lo_13_upstream = []
# csv_to_swc(csv_location, swc_directory, empty_list)
csv_to_swc(r"C:\Users\jjess\Documents\MEGA\Bellemer_L
# brain_graph(swc_list, colors)
brain_graph(plp_lo_13_upstream, "rainbow")
```

PLP.LO.13 upstream

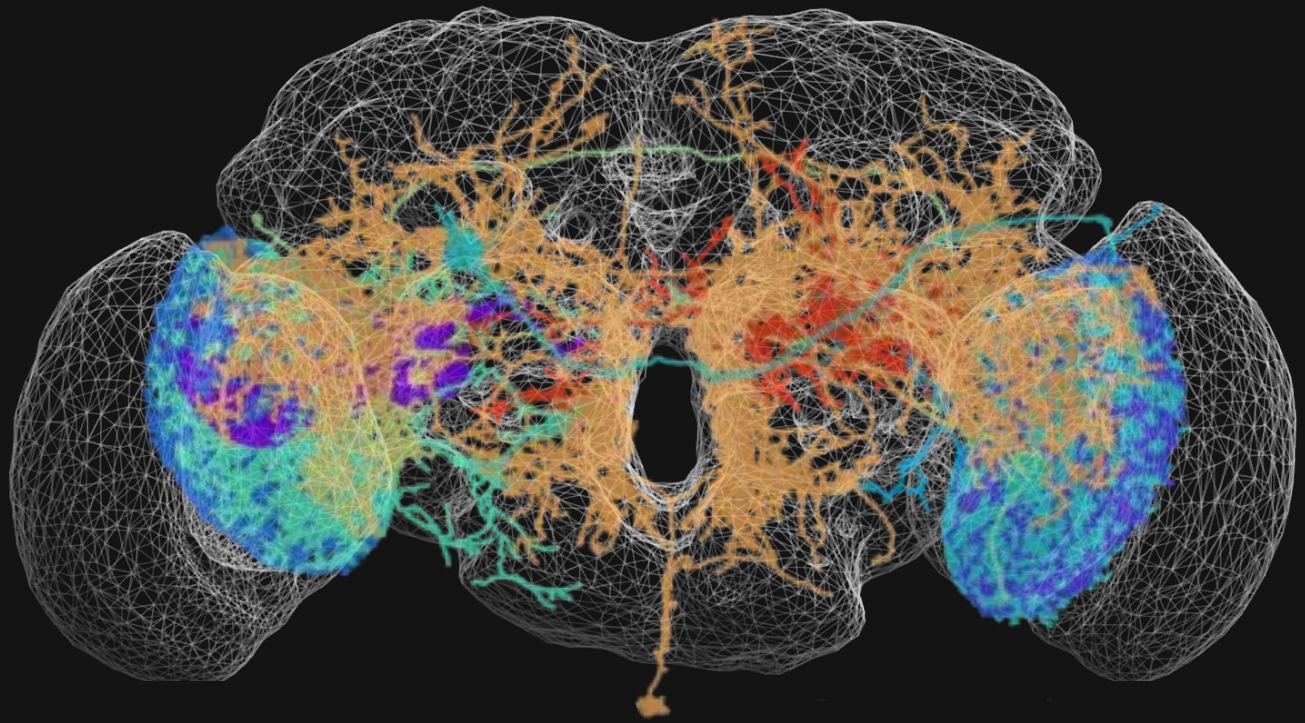
Neurons sending electrical activity into PLP.LO.13



```python
plp_lo_13_downstream = []
# csv_to_swc(csv_location, swc_directory, empty_list)
csv_to_swc(r"C:\Users\jjess\Documents\MEGA\Bellemer_Lab
# brain_graph(swc_list, colors)
brain_graph(plp_lo_13_downstream, "rainbow")
```

PLP.LO.13 downstream

Neurons receiving electrical activity from PLP.LO.13



My Approach

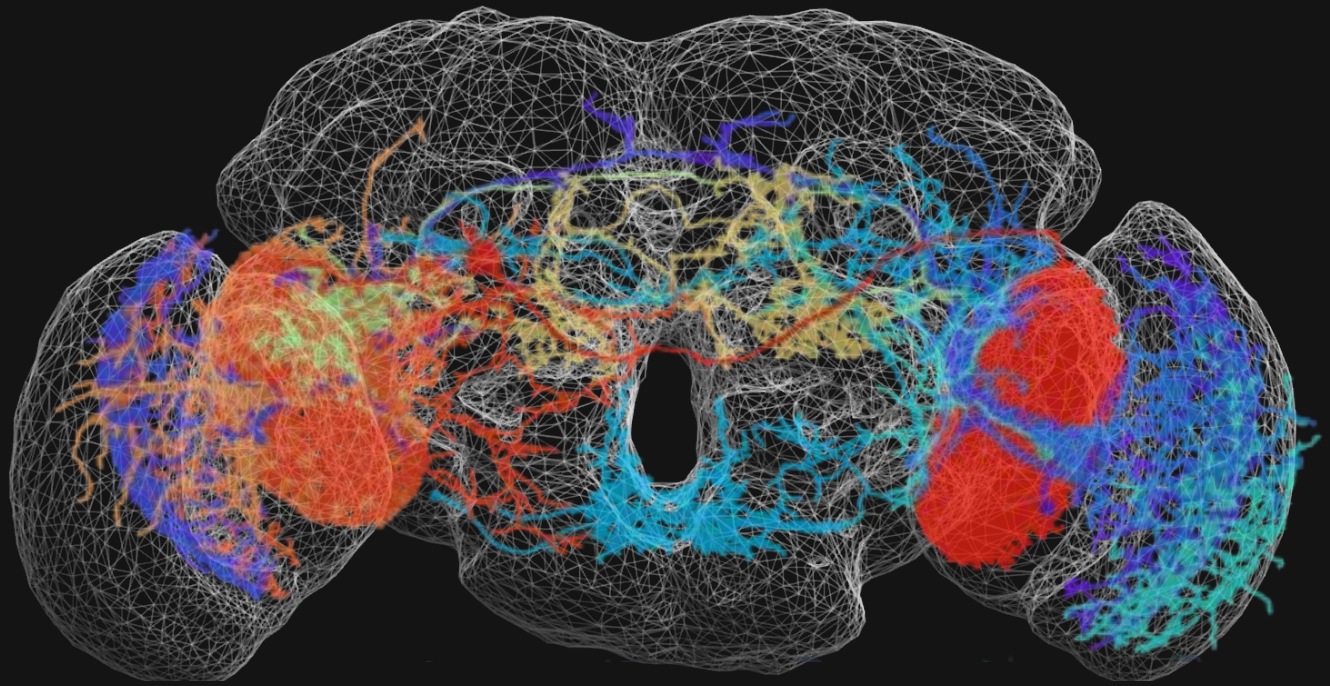
```python
def concatenator(csv_1_location, csv_2_location, csv_3_location=""):
    csv_1_file = pd.read_csv(csv_1_location, header = None)
    csv_2_file = pd.read_csv(csv_2_location, header = None)
    if csv_3_location != "":
        csv_3_file = pd.read_csv(csv_3_location, header = None)
        return pd.concat([csv_1_file, csv_2_file, csv_3_file], axis = 1)
    else:
        return pd.concat([csv_1_file, csv_2_file], axis = 1)

def shared_values(empty_list, df_1, df_2):
    for i in range(0, df_1.size):
        if (df_2.eq(df_1.iat[0,i]).any(axis = "columns")).loc[0]:
            empty_list.append(df_1.iat[0, i])

def shared_values_to_swc(shared_values_list, swc_directory, empty_list):
    for i in range(len(shared_values_list)):
        file_name = shared_values_list[i]
        swc_file_path =f"{swc_directory}\\{file_name}.swc"
        swc_file = navis.read_swc(swc_file_path)
        empty_list.append(swc_file)
    return empty_list
```

My Approach

```
plp_79_swc_concatenated = plp_79_up_list + plp_79_down_list + plp_79_reciprocal_list
plp_79_csv_concatenated = concatenator(r"C:\Users\jjess\Documents\MEGA\Bellemer_Lab\Pain_Connectome

plp_lo_13_swc_concatenated = plp_lo_13_up_list + plp_lo_13_down_list
plp_lo_13_csv_concatenated = concatenator(r"C:\Users\jjess\Documents\MEGA\Bellemer_Lab\Pain_Connect

allatostatin_c_connectome_list = []
allatostatin_c_connectome_swc_list = []

# shared_values(empty_list, df_1, df_2)
shared_values(allatostatin_c_connectome_list, plp_79_csv_concatenated, plp_lo_13_csv_concatenated)

# shared_values_to_swc(shared_values_list, swc_directory, empty_list)
shared_values_to_swc(allatostatin_c_connectome_list, r"C:\Users\jjess\Documents\MEGA\Bellemer_Lab\P

# brain_graph(swc_list, colors)
brain_graph(allatostatin_c_connectome_swc_list, "rainbow")
```

Shared 1st Order Neurons

Neurons sending, receiving, and reciprocating electrical activity between both PLP.79 and PLP.LO.13



Acknowledgements

Thank you.

Sources

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